A Appendix

A.1 Proof of Lemma 1

For a general Dirichlet distribution with parameter vector $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_n)$; $\alpha_i > 0$, the density function is given by $f(x_0, \ldots, x_n) = \dfrac{1}{B(\alpha)} \displaystyle\prod_{i=0}^{n} x_i^{\alpha_i - 1}$, where $x_i \geq 0$, $\displaystyle\sum_{i=0}^{n} x_i = 1$, and $B(\alpha) = \dfrac{\prod_{i=0}^{n} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=0}^{n} \alpha_i\right)}$ denotes the multivariate beta function. Using the standard properties of a Dirichlet distribution, one can obtain that $E[E_i] = \dfrac{\alpha_i}{\sum_{j=0}^{n} \alpha_j}$. Hence, under the Dirichlet$(1, 1, \ldots, 1)$ model, $E[E_i] = \dfrac{1}{n+1}$, for $i = 0, 1, \ldots, n$. Similarly, it is quite straightforward to obtain

$$E[E_i^2] = \frac{\alpha_i(\alpha_i + 1)}{\left(\sum_{j=0}^{n} \alpha_j\right)\left(\sum_{j=0}^{n} \alpha_j + 1\right)},$$

which yields $E[E_i^2] = \dfrac{2}{(n+1)(n+2)}$, $i = 0, 1, \ldots, n$. For $i \neq j$, as E_i and E_j are independent, one can derive that $E[E_i E_j] = E[E_i]\, E[E_j] = \dfrac{\alpha_i \alpha_j}{\left(\sum_{k=0}^{n} \alpha_k\right)\left(\sum_{k=0}^{n} \alpha_k + 1\right)}$, and therefore $E[E_i E_j] = \dfrac{1}{(n+1)(n+2)}$. Consequently, $\mathrm{Var}(E_i) = E[E_i^2] - \{E[E_i]\}^2 = \dfrac{n}{(n+1)^2(n+2)}$, while for $i \neq j$

$$\mathrm{Cov}(E_i, E_j) = E[E_i E_j] - E[E_i]E[E_j] = -\frac{1}{(n+1)^2(n+2)}.$$

Thus, $E(E_i) = \dfrac{1}{n+1}$, $\mathrm{Var}(E_i) = \dfrac{n}{(n+1)^2(n+2)}$, and, for $i \neq j$, $\mathrm{Cov}(E_i, E_j) = -\dfrac{1}{(n+1)^2(n+2)}$.

A.2 Proof of Theorem 1

Proof. From the definition of the proposed statistic, $T_{n,r}(w) = \dfrac{A_{n,r,w}}{B_{n,r}} + o_p(n^{-1/2})$, where the $o_p(n^{-1/2})$ term accounts for the negligible boundary contribution arising from the use of $n - r$ rather than n terms.

Define the function $g(a, b) = \dfrac{a}{b}$. Since $\mu_{B,r} > 0$, the function g is continuously differentiable in a neighborhood of $(\mu_{A,r,w}, \mu_{B,r})$. Its gradient is $\nabla g(a, b) = \begin{pmatrix} \frac{1}{b} \\ -\frac{a}{b^2} \end{pmatrix}$. Therefore, $\nabla g(\mu_{A,r,w}, \mu_{B,r}) = \begin{pmatrix} \frac{1}{\mu_{B,r}} \\ -\frac{\mu_{A,r,w}}{\mu_{B,r}^2} \end{pmatrix}$. By the assumed joint central limit theorem and the multivariate delta method,

$$\sqrt{n}\,\{g(A_{n,r,w}, B_{n,r}) - g(\mu_{A,r,w}, \mu_{B,r})\} \xrightarrow{d} N(0, \tau_{r,w}^2),$$

where $\tau_{r,w}^2 = \nabla g(\mu_{A,r,w}, \mu_{B,r})^\mathsf{T} \Sigma_{r,w} \nabla g(\mu_{A,r,w}, \mu_{B,r})$. Substituting the gradient and the covariance matrix gives

$$\tau_{r,w}^2 = \begin{pmatrix} \dfrac{1}{\mu_{B,r}} & -\dfrac{\mu_{A,r,w}}{\mu_{B,r}^2} \end{pmatrix} \begin{pmatrix} \sigma_{A,r,w}^2 & \sigma_{AB,r,w} \\ \sigma_{AB,r,w} & \sigma_{B,r}^2 \end{pmatrix} \begin{pmatrix} \dfrac{1}{\mu_{B,r}} \\ -\dfrac{\mu_{A,r,w}}{\mu_{B,r}^2} \end{pmatrix}$$

$$= \frac{\sigma_{A,r,w}^2}{\mu_{B,r}^2} + \frac{\mu_{A,r,w}^2 \sigma_{B,r}^2}{\mu_{B,r}^4} - \frac{2\mu_{A,r,w}\sigma_{AB,r,w}}{\mu_{B,r}^3}.$$

Moreover, $g(\mu_{A,r,w}, \mu_{B,r}) = \dfrac{\mu_{A,r,w}}{\mu_{B,r}} = \theta_{r,w}$. Hence, $\sqrt{n}\,(T_{n,r}(w) - \theta_{r,w}) \xrightarrow{d} N(0, \tau_{r,w}^2)$, which proves the result. □

A.3 Proof of Proposition 1

Proof. Under H_0, by the probability integral transform, we may work with $U_1, \dots, U_n \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0,1)$. Let

$$0 = U_{(0)} < U_{(1)} < \cdots < U_{(n)} < U_{(n+1)} = 1$$

and define the complete uniform spacings

$$E_j = U_{(j+1)} - U_{(j)}, \qquad j = 0, \dots, n.$$

Then

$$(E_0, \dots, E_n) \sim \text{Dirichlet}(1, \dots, 1).$$

Equivalently, there exist independent random variables $Y_0, \dots, Y_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$ such that

$$E_j = \frac{Y_j}{S_n}, \qquad S_n = \sum_{j=0}^{n} Y_j.$$

This representation is particularly convenient because the common normalizing factor S_n cancels from the ratio defining $T_{n,r}(w)$.

For a fixed admissible spacing order r, the r-th order spacing can be written as

$$D_i^{(r)} = U_{(i+r)} - U_{(i)} = \sum_{j=i}^{i+r-1} E_j = \frac{A_i^{(r)}}{S_n},$$

where

$$A_i^{(r)} = \sum_{j=i}^{i+r-1} Y_j, \qquad i = 1, \dots, n-r.$$

Moreover,

$$D_{i+1}^{(r)} - D_i^{(r)} = E_{i+r} - E_i = \frac{Y_{i+r} - Y_i}{S_n}.$$

Consequently, writing

$$N_{n,r}(w) = \sum_{i=1}^{n-r-1} w_{i,n} \left(D_{i+1}^{(r)} - D_i^{(r)} \right)^2$$

and

$$B_{n,r} = \sum_{i=1}^{n-r} \left(D_i^{(r)} - \overline{D}^{(r)} \right)^2,$$

we have

$$T_{n,r}(w) = \frac{N_{n,r}(w)}{B_{n,r}} = \frac{\sum_{i=1}^{n-r-1} w_{i,n}(Y_{i+r} - Y_i)^2}{\sum_{i=1}^{n-r} \left(A_i^{(r)} - \overline{A}^{(r)} \right)^2},$$

where

$$\overline{A}^{(r)} = \frac{1}{n-r} \sum_{i=1}^{n-r} A_i^{(r)}.$$

Thus, it remains to determine the probability limits of the numerator and denominator after division by n.

First consider the numerator. Define

$$Z_{i,r} = (Y_{i+r} - Y_i)^2.$$

For fixed r, the sequence $\{Z_{i,r}\}$ is stationary and has finite mean. Since Y_i and Y_{i+r} are independent exponential random variables with mean and variance equal to one,

$$E(Z_{i,r}) = E(Y_{i+r} - Y_i)^2 = \mathrm{Var}(Y_{i+r} - Y_i) + \{E(Y_{i+r} - Y_i)\}^2 = 2.$$

Because the sequence is generated from an independent sequence of exponential random variables and has only finite-range dependence, the law of large numbers for fixed-range dependent sequences gives

$$\frac{1}{n} \sum_{i=1}^{n-r-1} Z_{i,r} \xrightarrow{p} 2.$$

The boundedness and Riemann integrability of w imply that

$$\frac{1}{n} \sum_{i=1}^{n-r-1} w\left(\frac{i}{n}\right) Z_{i,r} \xrightarrow{p} 2 \int_0^1 w(x)\,dx.$$

Indeed, this follows from the weighted law of large numbers for finite-range dependent sequences together with the Riemann-sum convergence

$$\frac{1}{n} \sum_{i=1}^{n-r-1} w\left(\frac{i}{n}\right) \longrightarrow \int_0^1 w(x)\,dx.$$

Hence

$$\frac{1}{n} \sum_{i=1}^{n-r-1} w_{i,n}(Y_{i+r} - Y_i)^2 \xrightarrow{p} 2 \int_0^1 w(x)\,dx. \tag{A.1}$$

Next consider the denominator. For fixed r,

$$A_i^{(r)} = Y_i + \cdots + Y_{i+r-1}$$

is a stationary sequence with mean

$$E\{A_i^{(r)}\} = r$$

and variance

$$\mathrm{Var}\{A_i^{(r)}\} = r.$$

Furthermore, $\{A_i^{(r)}\}$ is $(r-1)$-dependent, since two such variables are independent whenever their defining blocks of exponentials do not overlap. Therefore, by the law of large numbers,

$$\overline{A}^{(r)} = \frac{1}{n-r}\sum_{i=1}^{n-r} A_i^{(r)} \xrightarrow{p} r,$$

and

$$\frac{1}{n-r}\sum_{i=1}^{n-r}\left(A_i^{(r)} - r\right)^2 \xrightarrow{p} \mathrm{Var}\{A_i^{(r)}\} = r.$$

Since $\overline{A}^{(r)} \xrightarrow{p} r$, the usual decomposition of the sample variance gives

$$\frac{1}{n-r}\sum_{i=1}^{n-r}\left(A_i^{(r)} - \overline{A}^{(r)}\right)^2 \xrightarrow{p} r.$$

Because r is fixed,

$$\frac{n-r}{n} \longrightarrow 1,$$

and hence

$$\frac{1}{n}\sum_{i=1}^{n-r}\left(A_i^{(r)} - \overline{A}^{(r)}\right)^2 \xrightarrow{p} r. \tag{A.2}$$

Combining (A.1) and (A.2), and using Slutsky's theorem,

$$T_{n,r}(w) = \frac{n^{-1}\displaystyle\sum_{i=1}^{n-r-1} w_{i,n}(Y_{i+r} - Y_i)^2}{n^{-1}\displaystyle\sum_{i=1}^{n-r}(A_i^{(r)} - \overline{A}^{(r)})^2} \xrightarrow{p} \frac{2\displaystyle\int_0^1 w(x)\,dx}{r}.$$

Therefore,

$$\boxed{T_{n,r}(w) \xrightarrow{p} \frac{2}{r}\int_0^1 w(x)\,dx = \theta_{r,w}}.$$

This proves the proposition. □

A.4 Proof of Theorems 2 and 3

Proof. We prove the two results jointly by first establishing the joint asymptotic behavior of the numerator and denominator of $T_{n,r}(w)$ and then applying the multivariate delta method. The proof under the contiguous alternatives follows from the same joint central limit argument after incorporating the corresponding local mean shift.

Under H_0, by the probability integral transform, we may assume without loss of generality that

$$U_1, \ldots, U_n \overset{\text{i.i.d.}}{\sim} \mathrm{Uniform}(0,1).$$

Let

$$0 = U_{(0)} < U_{(1)} < \cdots < U_{(n)} < U_{(n+1)} = 1$$

and define the elementary spacings

$$E_j = U_{(j+1)} - U_{(j)}, \qquad j = 0, \ldots, n.$$

The vector of spacings has the Dirichlet representation

$$(E_0, \dots, E_n) \overset{d}{=} \left(\frac{Y_0}{S_n}, \dots, \frac{Y_n}{S_n} \right), \qquad S_n = \sum_{j=0}^{n} Y_j,$$

where $Y_0, \dots, Y_n$ are independent Exp(1) random variables. For a fixed admissible order r,

$$D_i^{(r)} = U_{(i+r)} - U_{(i)} = \frac{A_i^{(r)}}{S_n}, \qquad A_i^{(r)} = \sum_{j=i}^{i+r-1} Y_j.$$

Moreover,

$$D_{i+1}^{(r)} - D_i^{(r)} = \frac{Y_{i+r} - Y_i}{S_n}.$$

Write

$$T_{n,r}(w) = \frac{N_{n,r}(w)}{Q_{n,r}},$$

where

$$N_{n,r}(w) = \sum_{i=1}^{n-r-1} w_{i,n} \left(D_{i+1}^{(r)} - D_i^{(r)} \right)^2$$

and

$$Q_{n,r} = \sum_{i=1}^{n-r} \left(D_i^{(r)} - \overline{D}^{(r)} \right)^2.$$

Since the same factor S_n^{-2} occurs in both components, it cancels from the ratio. Thus,

$$T_{n,r}(w) = \frac{\sum_{i=1}^{n-r-1} w_{i,n}(Y_{i+r} - Y_i)^2}{\sum_{i=1}^{n-r} \left(A_i^{(r)} - \overline{A}^{(r)} \right)^2},$$

where

$$\overline{A}^{(r)} = \frac{1}{n-r} \sum_{i=1}^{n-r} A_i^{(r)}.$$

Define

$$X_{i,r} = (Y_{i+r} - Y_i)^2$$

and

$$V_{i,r} = \left(A_i^{(r)} - r \right)^2.$$

For fixed r, both sequences are stationary finite-range dependent sequences with finite moments of all orders. Hence the corresponding central limit theorems for fixed-range dependent triangular arrays apply.

Since Y_i and Y_{i+r} are independent Exp(1) variables,

$$E(X_{i,r}) = E(Y_{i+r} - Y_i)^2 = 2.$$

Consequently, using $w_{i,n} = w(i/n)$,

$$\frac{1}{n} \sum_{i=1}^{n-r-1} w_{i,n} X_{i,r} \xrightarrow{p} 2 \int_0^1 w(x)\, dx.$$

Furthermore, the weighted central limit theorem yields

$$\sqrt{n}\left\{\frac{1}{n}\sum_{i=1}^{n-r-1} w_{i,n}X_{i,r} - 2\int_0^1 w(x)\,dx\right\} \xrightarrow{d} N(0,\tau^2_{11,r,w}), \tag{A.3}$$

for a finite asymptotic variance $\tau^2_{11,r,w}$ determined by the long-run covariance structure of the finite-range dependent sequence $\{w_{i,n}X_{i,r}\}$.

For the denominator, observe that

$$A_i^{(r)} = Y_i + \cdots + Y_{i+r-1},$$

so that

$$E\{A_i^{(r)}\} = r, \qquad \mathrm{Var}\{A_i^{(r)}\} = r.$$

Since the sequence $\{A_i^{(r)}\}$ is $(r-1)$-dependent,

$$\overline{A}^{(r)} \xrightarrow{p} r$$

and

$$\frac{1}{n}\sum_{i=1}^{n-r}\left(A_i^{(r)} - \overline{A}^{(r)}\right)^2 \xrightarrow{p} r.$$

The corresponding sample-variance central limit theorem therefore gives

$$\sqrt{n}\left\{\frac{1}{n}\sum_{i=1}^{n-r}\left(A_i^{(r)} - \overline{A}^{(r)}\right)^2 - r\right\} \xrightarrow{d} N(0,\tau^2_{22,r}). \tag{A.4}$$

The two components are constructed from the same sequence of exponential variables and are consequently not, in general, asymptotically independent. Their joint central limit theorem can be written as

$$\sqrt{n}\begin{pmatrix} N^*_{n,r} - 2W \\ Q^*_{n,r} - r \end{pmatrix} \xrightarrow{d} N_2\left[\begin{pmatrix}0\\0\end{pmatrix}, \Sigma_{r,w}\right], \tag{A.5}$$

where

$$W = \int_0^1 w(x)\,dx,$$

$$N^*_{n,r} = \frac{1}{n}\sum_{i=1}^{n-r-1} w_{i,n}(Y_{i+r} - Y_i)^2,$$

$$Q^*_{n,r} = \frac{1}{n}\sum_{i=1}^{n-r}\left(A_i^{(r)} - \overline{A}^{(r)}\right)^2,$$

and

$$\Sigma_{r,w} = \begin{pmatrix} \tau^2_{11,r,w} & \tau_{12,r,w} \\ \tau_{12,r,w} & \tau^2_{22,r} \end{pmatrix}.$$

The entries of $\Sigma_{r,w}$ are the corresponding long-run covariances. Their explicit forms are finite because r is fixed and the exponential variables possess finite moments of every order.

Now consider the smooth map

$$g(a,b) = \frac{a}{b}.$$

At the probability limit

$$(a_0, b_0) = (2W, r),$$

its gradient is

$$\nabla g(a_0, b_0) = \left(\frac{1}{r}, -\frac{2W}{r^2}\right)^\top.$$

Since

$$\mu_{r,w} = g(2W, r) = \frac{2}{r}\int_0^1 w(x)\,dx,$$

the multivariate delta method applied to (A.5) gives

$$\sqrt{n}\,\{T_{n,r}(w) - \mu_{r,w}\} \xrightarrow{d} N(0, \sigma^2_{r,w}),$$

where

$$\boxed{\sigma^2_{r,w} = \nabla g(2W, r)^\top \Sigma_{r,w} \nabla g(2W, r)}.$$

Equivalently,

$$\sigma^2_{r,w} = \frac{\tau^2_{11,r,w}}{r^2} + \frac{4W^2\tau^2_{22,r}}{r^4} - \frac{4W\tau_{12,r,w}}{r^3}.$$

Under the stated non-degeneracy conditions, $\sigma^2_{r,w} > 0$. This proves Theorem 2.

We next consider the sequence of contiguous alternatives $H_{1,n}$. Let the corresponding densities be written as

$$f_n(x) = f_0(x)\left\{1 + \frac{c}{\sqrt{n}} h(x)\right\},$$

where h satisfies the normalization and regularity conditions imposed for the contiguous-alternative analysis. Under $H_{1,n}$, the same joint central limit argument applies, but the numerator and denominator acquire deterministic first-order shifts induced by the perturbation h.

More precisely, suppose that the joint expansion can be written as

$$\sqrt{n}\begin{pmatrix} N^*_{n,r} - 2W \\ Q^*_{n,r} - r \end{pmatrix} \xrightarrow{d} N_2\left[\begin{pmatrix} \eta_{1,r,w}(h) \\ \eta_{2,r}(h) \end{pmatrix}, \Sigma_{r,w}\right]. \tag{A.6}$$

The quantities $\eta_{1,r,w}(h)$ and $\eta_{2,r}(h)$ denote the first-order changes in the numerator and denominator generated by the local perturbation. Their precise forms depend on the perturbation function h, the spacing order r, and the weighting function w.

Applying the same delta-method expansion at $(2W, r)$ gives

$$\sqrt{n}\,\{T_{n,r}(w) - \mu_{r,w}\} \xrightarrow{d} N\left(\Delta_h, \sigma^2_{r,w}\right),$$

where

$$\boxed{\Delta_h = \nabla g(2W, r)^\top \begin{pmatrix} \eta_{1,r,w}(h) \\ \eta_{2,r}(h) \end{pmatrix}}$$

and hence

$$\boxed{\Delta_h = \frac{\eta_{1,r,w}(h)}{r} - \frac{2W}{r^2}\eta_{2,r}(h).}$$

Thus, the local mean shift is determined by the interaction between the perturbation h, the spacing order r, and the weighting scheme w.

In the notation of the local asymptotic analysis, this first-order shift may equivalently be represented through the sensitivity coefficient $\kappa_{r,w}(h)$, so that

$$\Delta_h = c\,\kappa_{r,w}(h).$$

Therefore, whenever $\kappa_{r,w}(h) \neq 0$, the contiguous alternative produces a non-zero displacement of the limiting normal distribution and hence a non-trivial local asymptotic power function. In contrast, if

$$\kappa_{r,w}(h) = 0,$$

the first-order local displacement vanishes and the statistic has the same first-order limiting distribution as under the null. In such cases, the corresponding $n^{-1/2}$ alternatives are not detected at the first-order level, which motivates consideration of alternative detection rates.

The covariance matrix in (A.6) has the same first-order limiting structure as under the null under the stated contiguity and regularity conditions. Consequently, the asymptotic variance remains $\sigma^2_{r,w}$, while the contiguous alternative affects the limiting distribution through the mean shift Δ_h. This establishes Theorem 3 and completes the joint proof. $\square$